

Table 1: Planned future-state evaluation across four dimensions. All results are pending; arrows indicate the preferred direction. ZS denotes zero-shot inference with a public checkpoint.

Method	Future clinical status		Disease progression		Future report fidelity		Probabilistic reliability	
	AP ↑	AUROC ↑	Trans. F1↑	Dir. F1↑	RadGraph F1↑	GREEN ↑	Brier ↓	ECE ↓
Qwen2.5-VL-7B ^{ZS}	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
LLaVA-v1.6-Mistral-7B ^{ZS}	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
LLaVA-Med-v1.5-7B ^{ZS}	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
CheXagent-8B ^{ZS}	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
BioViL-T + predictor	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
CheXWorld + predictor	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
Copy Current	—	—	TBD	TBD	TBD	TBD	—	—
Finding-transition prior	TBD	TBD	TBD	—	—	—	TBD	TBD
Qwen3.5-0.8B direct	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
MedWorld-JEPA (Ours)	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD

AP is non-interpolated average precision; AP, AUROC, Brier and classwise ECE are macro-averaged over findings. Trans. F1 covers onset/resolution; Dir. F1 covers improved/stable/worsened persistent findings. RadGraph uses partial F1. VLM scores use Yes/No likelihoods. “+ predictor” denotes trained fusion, prediction and readouts. Copy Current has no continuous-score interface; the finding prior has no direction or report output. Direct and Ours share a 0.8B language backbone and forecast-training data. TBD: pending; —: not evaluated.

Table 1 plan: future prediction

Predict the unobserved follow-up from the current radiograph, current report, source-available clinical history, and requested horizon. Ten methods/settings are evaluated across four dimensions, with two metrics each; zero-shot rows are marked explicitly.

Planned scoring interface. For each finding, score the fixed candidate answers *Yes* and *No* from the VLM’s conditional token likelihoods:

$$s_k = \frac{\exp(\ell_{\text{Yes},k})}{\exp(\ell_{\text{Yes},k}) + \exp(\ell_{\text{No},k})}.$$

Here ℓ is the candidate answer’s log-likelihood, scored at the same answer boundary. No extra classification head is required for this interface. Representation forecasters use supervised finding heads; the interface and training regime will be disclosed per row. AP/AUROC assess ranking; Brier/ECE assess the same scores as probability predictions. The main table uses raw scores; any calibration uses validation data only and is reported separately.

Evaluation plan. All rows share patient splits, per-metric reference masks, and clinical scorers. Transition F1 averages supported finding-specific onset/resolution events. Direction F1 measures improved/stable/worsened persistent findings using reviewed, source-anchored annotations; stable references remain eligible for false changes. Report fidelity uses RadGraph partial F1 and GREEN. Brier and ten-bin classwise ECE are macro-averaged over findings. Onset/resolution breakdowns, retrieval, CheXbert F1, horizon analyses, coverage, and patient-bootstrap intervals are supplementary. Direction-label validation and new scorers remain pending.

Table 2: Planned comparison on six downstream tasks. All results are pending; higher is better. ZS denotes no benchmark-specific adaptation.

Method	Classification		VQA		Report gen.		Grounding		Segmentation		SR 4×	
	AUC	AP	Acc.	μ F1	RG	CB	mIoU	A@.5	Dice pseudo	Dice human	PSNR	SSIM
Qwen2.5-VL-7B ^{ZS}	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	—	—	—	—
LLaVA-v1.6-7B ^{ZS}	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	—	—	—	—
LLaVA-Med-v1.5-7B ^{ZS}	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	—	—	—	—
CheXagent-8B ^{ZS}	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	—	—	—	—
DINOv2 + heads	TBD	TBD	—	—	—	—	—	—	TBD	TBD	TBD	TBD
CheXWorld + heads	TBD	TBD	—	—	—	—	—	—	TBD	TBD	TBD	TBD
MAIRA-2	—	—	—	—	TBD	TBD	TBD	TBD	—	—	—	—
SwinIR (CXR-adapted)	—	—	—	—	—	—	—	—	—	—	TBD	TBD
Qwen3.5-0.8B no-slots	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
MedWorld-JEPA (Ours)	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD

TBD: pending; —: not evaluated with this interface. DINOv2/CheXWorld use frozen encoders and supervised heads; MAIRA-2 uses its native reporting/grounding interface. The no-slots baseline and Ours use matched six-task supervision and decoders. RG/CB: RadGraph/CheXbert F1; μ F1: micro F1; A@.5: box accuracy at IoU ≥ 0.5 . Pseudo/human Dice are separate tests. Grounding conversion and matching, human-mask data, and pretraining overlap will be specified before final evaluation.

Table 2 plan: six downstream tasks

Evaluate current-image understanding, language readout, localization, and reconstruction. The six task families are classification, standard VQA, current report generation, phrase grounding, organ segmentation, and spatial 4× super-resolution.

Data. Use MIMIC-CXR for classification, reports and synthetic SR; MIMIC-CXR-VQA for standard question answering; MS-CXR for phrase grounding; and CXAS pseudo-labels plus a separate human-mask test for segmentation. Classification uses macro AUROC/AP; reports use RadGraph partial and CheXbert macro-positive F1. Spatial 4× SR has 1/16 of the HR pixel count.

Input and training plan. Use images plus task questions/phrases and withhold the same-exam target report. Adapt Ours and no-slots to this setting; exclude test patients across all training tasks and stages. Report VQA Verify/Choose/Query and whole-image/region slices separately. Added readouts require training; current report-assisted four-task scores are not entries in this plan.